

# Week 4 : 1D Euler equation

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# 1D Euler Equations

## 1D Euler Equation

$$\frac{\partial \vec{Q}}{\partial t} + \frac{\partial \vec{E}}{\partial x} = 0$$

With the conserved quantities  $\vec{Q} = [\rho, \rho u, E_t]$  and the fluxes

$$\vec{E} = [\rho u, \rho u^2 + P, (E_t + p)u]$$

For a perfect gaz :

$$P = \rho R T$$

$$e = c_v T$$

$$h = c_p T$$

$$c_v = \frac{R}{\gamma - 1}$$

$$c_p = \frac{R\gamma}{\gamma - 1}$$

$$H = E + \frac{P}{\rho}$$

$$E = e + \frac{1}{2}u^2$$

$$E_t = \rho \left( e + \frac{1}{2}u^2 \right)$$

$$c = \sqrt{\gamma R T}$$

- Density :  $\rho$  ;
- Speed :  $u$  ;
- Momentum :  $\rho u$  ;
- Total massic energy :  $E$  ;
- Enthalpy :  $H$
- Pressure :  $P$
- Speed of sound :  $c$
- Temperature :  $T$

## Jacobian Matrix

$$A = \frac{\partial \vec{E}}{\partial \vec{Q}} = \begin{bmatrix} 0 & 1 & 0 \\ (\gamma - 3)\frac{u^2}{2} & (3 - \gamma)u & \gamma - 1 \\ (\gamma - 1)u^3 - \frac{\gamma u E_t}{\rho} & -\frac{3}{2}(\gamma - 1)u^2 + \frac{\gamma E_t}{\rho} & \gamma u \end{bmatrix}$$

Eigenvalues of the Jacobian matrix :  $|A - \lambda I| = 0$

$$\begin{aligned} & -\lambda \left( [(3 - \gamma) - \lambda](\gamma u - \lambda) \right) - (\gamma - 1) \left[ \frac{-3}{2}(\gamma - 1)u^2 + \gamma E_t / \rho \right] \\ & - \left( (\gamma - 3)\frac{u^2}{2}(\gamma u - \lambda) - (\gamma - 1) \left[ (\gamma - 1)u^3 - \gamma u E_t / \rho \right] \right) = 0 \end{aligned}$$

Three solution which represents the wave propagation speed in a fluid :

$$\lambda_1 = u$$

$$\lambda_2 = u + c$$

$$\lambda_3 = u - c$$



# Boundary conditions

Supersonic inlet (extrapolation from upstream)

$$\rho_1 = \rho_\infty$$

$$u_1 = u_\infty$$

$$P_1 = P_\infty$$

Supersonic outlet (extrapolation from upstream)

$$\rho_N = \rho_{N-1}$$

$$u_N = u_{N-1}$$

$$P_N = P_{N-1}$$

Subsonic outlet (Riemann invariants meeting at the boundary)

$$s = \text{cte along } u$$

$$u+2c/(\gamma-1) = R_1 \text{ along } u+c, \text{right running}$$

$$u-2c/(\gamma-1) = R_2 \text{ along } u-c, \text{left running}$$

The flow is isentropic at the exit so assuming the area to be constant we can compute the outside density with isentropic relation. We can compute the right running invariant from the inside. We know the back pressure  $P_L$  along the left running invariant. Thus we can compute  $c_L$  and  $\rho_L$  with isentropic relations. We compute  $R_1$  from the interior state, then  $R_2$  and  $u_L$  as :

$$R_2 = R_1 - c_L/(\gamma - 1)$$

$$u_L = (R_1 + R_2)/2$$

# 1D Mac Cormack Method

$$\vec{Q} = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix}$$

$$\vec{E} = \begin{bmatrix} \rho u \\ (\rho u^2 + p) \\ \rho u H \end{bmatrix}$$

Predictor step (Backward Difference) :

$$\vec{\bar{Q}}_i = \vec{Q}_i^n - \Delta t \left( \frac{\vec{E}_i^n - \vec{E}_{i-1}^n}{\Delta x} \right)$$

Corrector step (Forward Difference) :

$$\vec{\bar{\bar{Q}}}_i = \vec{Q}_i^n - \Delta t \left( \frac{\vec{\bar{E}}_{i+1}^n - \vec{\bar{E}}_i^n}{\Delta x} \right)$$

$\vec{\bar{E}}$  is computed from the vector  $\vec{\bar{Q}}$

$$\vec{Q}_i^{n+1} = \frac{1}{2} \left( \vec{\bar{Q}}_i + \vec{\bar{\bar{Q}}}_i \right)$$

In some case, doing the predictor step with a forward difference and the corrector step by a backward difference work best.

# Quasi-1D Euler equations

## 1D Euler Equation

$$\frac{\partial \vec{Q}}{\partial t} + \frac{\partial \vec{E}}{\partial x} = \vec{S}$$

With the conserved quantities  $\vec{Q} = [\rho A, \rho u A, \rho E A]$ , the fluxes

$\vec{E} = [\rho u A, (\rho u^2 + P) A, \rho u H A]$  and the source terms  $\vec{S} = [0, P \frac{dA}{dx}, 0]$

For a perfect gaz :

$$P = \rho R T$$

$$e = c_v T$$

$$h = c_p T$$

$$c_v = \frac{R}{\gamma - 1}$$

$$c_p = \frac{R\gamma}{\gamma - 1}$$

$$H = E + \frac{P}{\rho}$$

$$E = e + \frac{1}{2} u^2$$

$$E_t = \rho \left( e + \frac{1}{2} u^2 \right)$$

$$c = \sqrt{\gamma R T}$$

- Density :  $\rho$  ;
- Speed :  $u$  ;
- Momentum :  $\rho u$  ;
- Total massic energy :  $E$  ;
- Enthalpy :  $H$
- Pressure :  $P$
- Speed of sound :  $c$
- Temperature :  $T$
- Area of the nozzle :  $A(x)$

# Quasi-1D Mac Cormack Method

$$\vec{Q} = \begin{bmatrix} \rho A \\ \rho u A \\ \rho E A \end{bmatrix} \quad \vec{E} = \begin{bmatrix} \rho u A \\ (\rho u^2 + p) A \\ \rho u H A \end{bmatrix} \quad \vec{S} = \begin{bmatrix} 0 \\ p \frac{dA}{dx} \\ 0 \end{bmatrix}$$

Predictor step :

$$\bar{\bar{Q}}_i = Q_i^n - \Delta t \left( \frac{\vec{E}_i^n - \vec{E}_{i-1}^n}{\Delta x} \right) + \Delta t \vec{S}_i^n = 0$$

Corrector step :

$$\bar{\bar{Q}}_i = Q_i^n - \Delta t \left( \frac{\vec{E}_{i+1}^n - \vec{E}_i^n}{\Delta x} \right) + \Delta t \vec{S}_i^n$$

$\bar{\bar{E}}$  is computed from the vector  $\bar{\bar{Q}}$

$$\vec{Q}_i^{n+1} = \frac{1}{2} \left( \bar{\bar{Q}}_i + \bar{\bar{Q}}_i \right)$$



# Stability

For explicit scheme we need to have the CFL number lower than unity (like for the Lax scheme for the wave equation). This means :

$$u\Delta t/\Delta x \leq 1 \text{ for } \lambda = u$$

$$(u + c)\Delta t/\Delta x < 1 \text{ for } \lambda = u + c$$

$$(u - c)\Delta t/\Delta x < 1 \text{ for } \lambda = u - c$$

Thus to be conservative :

$$CFL = (|u| + c)\Delta t/\Delta x < 1$$



# Convergence monitoring

For steady-state problems, we integrate in time until the solution no longer changes. We can define a residual as :

$$|u^{n+1} - u^n|_2 = \sqrt{\sum_j (u_j^{n+1} - u_j^n)^2}$$

where  $u$  is any quantities, usually we look at the conserved quantities ( $\rho$ ,  $\rho u$ ,  $\rho v$ ,  $\rho E$ ).

